PH45130

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Quantum Gravitation

Lecture 13

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Propagation through the Cosmic Medium

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$D_T^{-1} = D_0^{-1} - \Sigma_T(\omega, k)$$

Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$k^2(z) = k_0^2 + \Pi_R(z) + i\Pi_I(z)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$S_N = 2coth(\hbar\omega/2k_BT_m)Im\Sigma_T$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 13; Cosmic Medium Propagation Tables; 66-Parameter cross-alias map